

LECTURE 1: SEGMENTATION

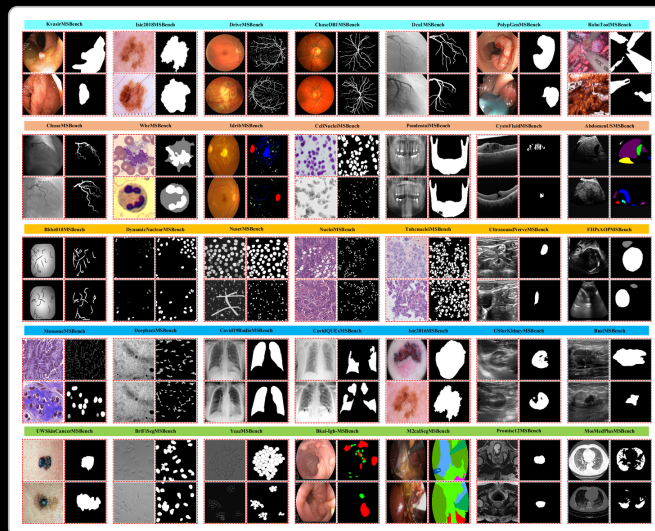
FROM U-NET TO nnU-NET: ONE LABEL PER PIXEL

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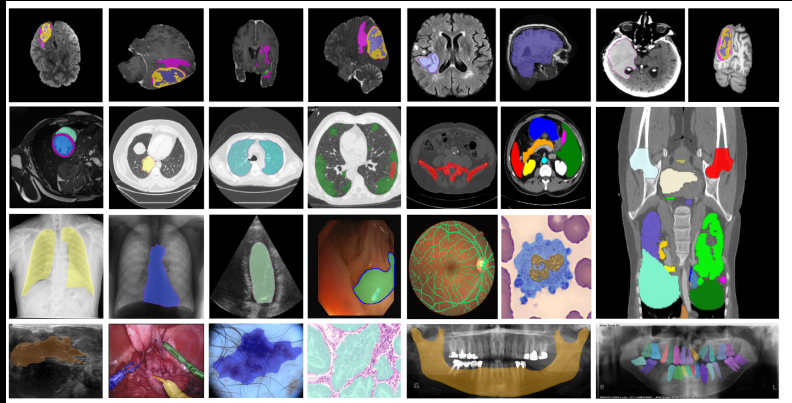
August 31, 2026

SEGMENTATION IS A DENSE CLASSIFICATION TASK

- ▶ Classification: **one label per image.**
- ▶ Segmentation: **one label per pixel.**



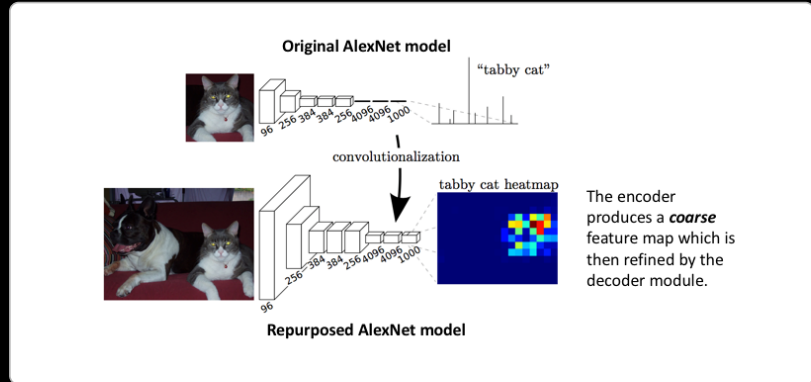
WHY CLINICIANS WANT MASKS, NOT LABELS



Medical image segmentation is used for **diagnosis**, **treatment planning**, and **patient care**:
because a label cannot be measured, and a mask can.

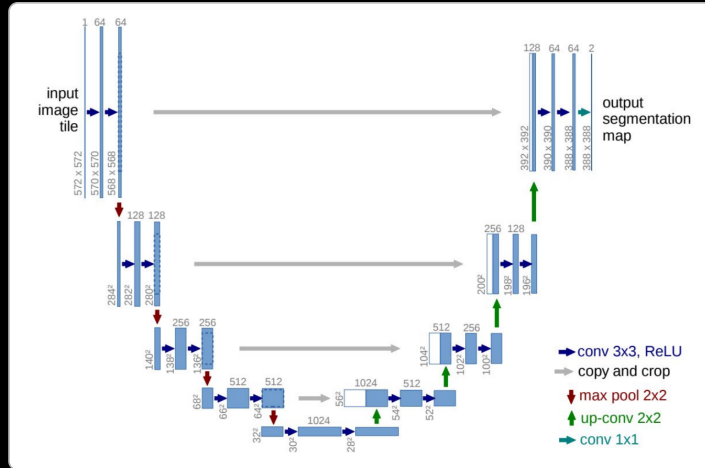
The U-Net

THE RESOLUTION PROBLEM



Segmentation needs a **full-resolution** output.

THE U-NET



Contracting path

Two 3×3 **unpadded convolutions** + ReLU, then 2×2 **max pooling**, stride 2. Each step **doubles the feature channels**.

Expansive path

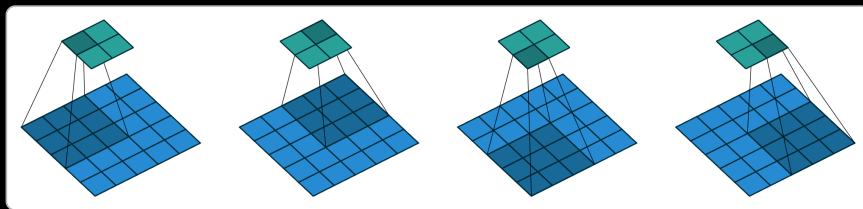
Upsampling, then a 2×2 **up-convolution** that halves the **feature channels**, and two 3×3 convolutions + ReLU.

Copy and crop

Concatenation with the **correspondingly cropped** feature map from the contracting path: every unpadded convolution loses border pixels.

CONVOLUTION (2D)

A 3×3 kernel over a 5×5 input, stride 2: output 2×2 .

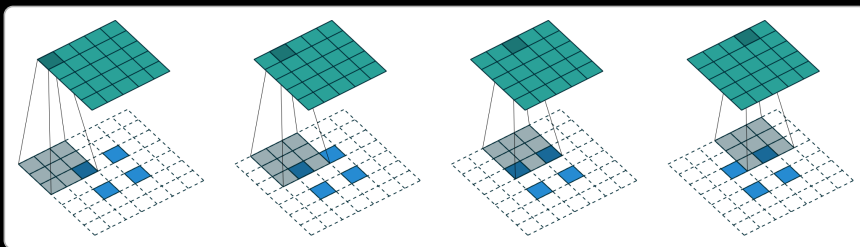


$$y[r, c] = \sum_{m=0}^{k-1} \sum_{n=0}^{k-1} x[sr + m, sc + n] w[m, n]$$

$$\text{output size } o = \left\lfloor \frac{i + 2p - k}{s} \right\rfloor + 1 = \left\lfloor \frac{5 + 0 - 3}{2} \right\rfloor + 1 = 2$$

TRANPOSED CONVOLUTION

The transpose of that convolution: 2×2 in, 5×5 out.



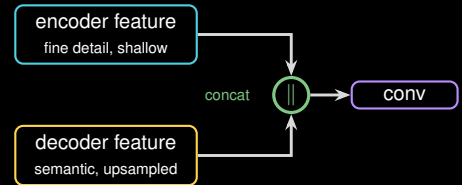
$$y[sr + m, sc + n] += x[r, c] w[m, n]$$

$$\text{output size } o = s(i - 1) + k - 2p = 2(2 - 1) + 3 - 0 = 5$$

$$y = C^T x: \text{ hence } \textit{transposed}.$$

SKIP CONNECTIONS: WHERE THE BOUNDARIES COME FROM

- ▶ Upsample the coarse decoder feature, then **concatenate** the matching encoder feature map.
- ▶ The skip connection is designed to help transmit the **spatial information** that gets lost during the encoder stages, to help recover the **high-resolution information**.
- ▶ Result: semantics from the deep path, **precise boundaries** from the shallow one.



How do we know it worked?

IoU (JACCARD INDEX)

Intersection over union

$$\text{IoU} = \frac{TP}{TP + FP + FN}$$

Pixels in *both* masks, over pixels in *either*. Range [0, 1], higher is better.

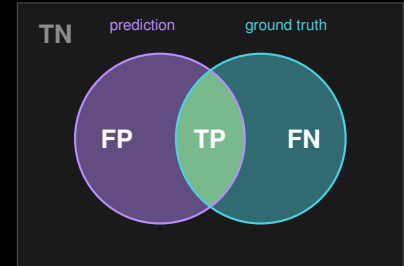
Work one through

Prediction 100 px, ground truth 120 px, overlap 80 px.

$TP=80$, $FP=20$, $FN=40$.

$\text{IoU} = 80/140 = \mathbf{0.57}$ $\text{Dice} = 160/220 = \mathbf{0.73}$

Compute **per image**, then average. Do not pool over the test set: pooling lets big easy cases hide small hard ones.



TN never appears, which is exactly why IoU survives class imbalance and accuracy does not.

DICE COEFFICIENT

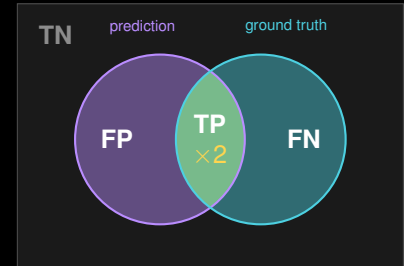
The metric medical imaging reports

$$\text{DSC} = \frac{2 TP}{2 TP + FP + FN}$$

Twice the intersection, over the sum of the two mask sizes, and again, no *TN*.

Also known as

The **F1 score** computed over pixels: precision = $TP/(TP+FP)$ and recall = $TP/(TP+FN)$, harmonic mean. Also called DSC, the Dice similarity coefficient.



TP is counted **twice** in the numerator: that is the whole difference from IoU.

DICE VS. IOU: NEVER ARGUE ABOUT WHICH IS BETTER

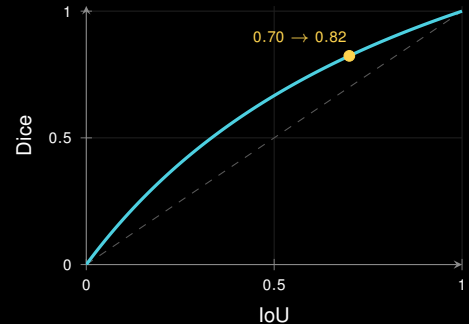
They are the same ranking

$$\text{Dice} = \frac{2 \text{IoU}}{1 + \text{IoU}} \quad \text{IoU} = \frac{\text{Dice}}{2 - \text{Dice}}$$

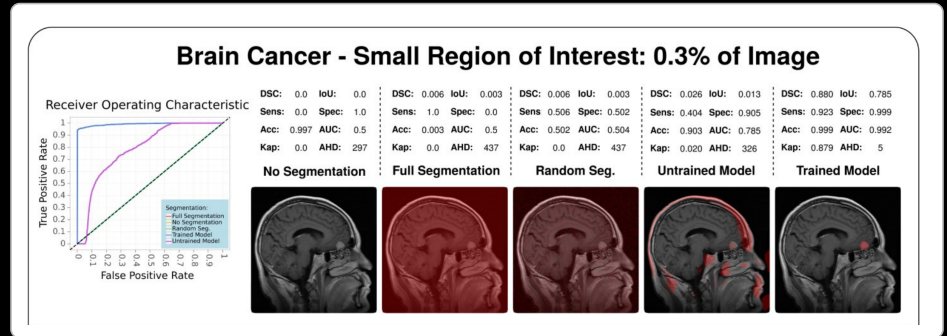
A strictly increasing map. If model A beats B on Dice, it beats B on IoU.

But the numbers differ

Dice is **never smaller** than IoU, and strictly larger whenever the masks **partially overlap**. IoU 0.70 is Dice 0.82.



ACCURACY IS HIGHLY DISCOURAGED

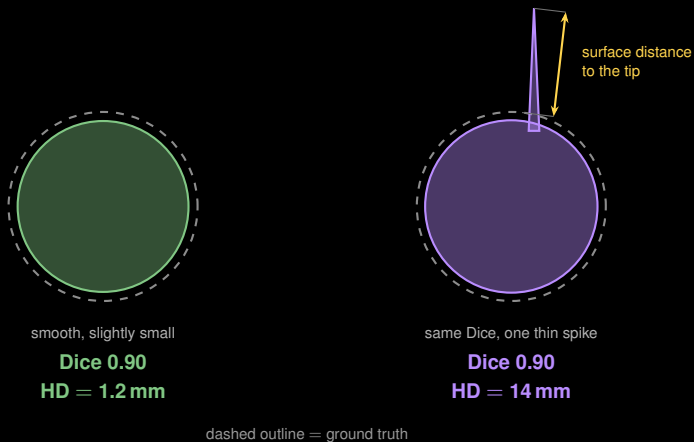


From finding **nothing** to finding the tumour, accuracy moves **0.997** to **0.999**. Dice moves **0.000** to **0.880**.

Dice and IoU are both appropriate; accuracy is not.

OVERLAP IS NOT ENOUGH

Dice and IoU measure **overlap**, **not boundary accuracy**. Two predictions, **identical Dice**:



The spike is **2.4% of the area** but **31% of the boundary**: Dice barely moves, the boundary measure explodes.

Both panels are constructed to score Dice 0.90 against the same ground truth. Maier-Hein et al., *Metrics Reloaded*, Nature Methods

HAUSDORFF DISTANCE AND HD95

Distance-based measures score the **spatial distance between predicted and ground-truth boundaries**.

$$HD = \max \left(\sup_{p \in P} \inf_{g \in G} d(p, g), \sup_{g \in G} \inf_{p \in P} d(g, p) \right)$$

the maximum distance from a boundary point of one set to the closest point of the other: the **worst-case** boundary error

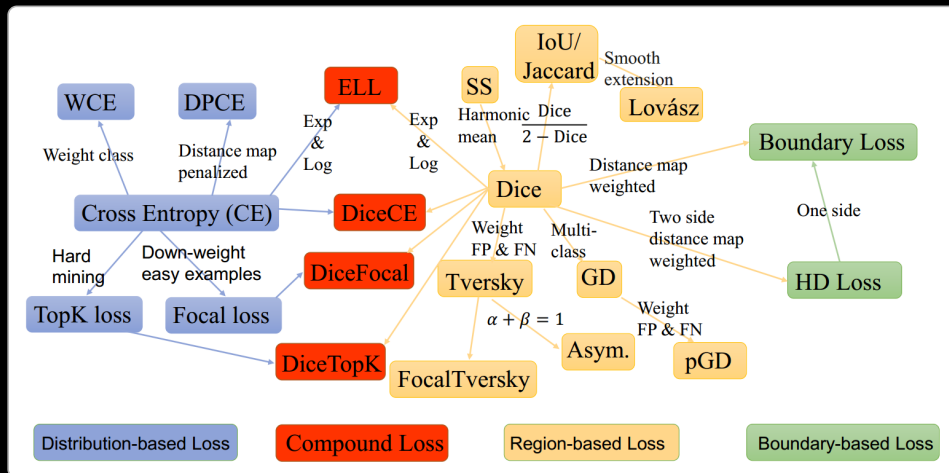
HD95: the maximum is susceptible to outliers, so report the **95th percentile** instead.

It is measured in millimetres: resample to a common voxel spacing before comparing.

P , G are the predicted and ground-truth boundary point sets. HD: Hausdorff. HD95: Huttenlocher, Klanderman & Rucklidge, IEEE TPAMI; standardised by Taha & Hanbury, BMC Medical Imaging. HD95 is the standard boundary metric in BraTS and the Decathlon.

Training it

LOSS FUNCTION FOR MEDICAL IMAGE SEGMENTATION



Distribution-based, region-based, boundary-based, and the compound losses that mix them.

CROSS-ENTROPY VS. DICE LOSS

Pixelwise cross-entropy

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C g_i^c \log s_i^c$$

Treats every pixel as an independent classification: g_i^c is the binary indicator that class c is correct at pixel i , and s_i^c is the predicted probability.

Dice loss

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}$$

Optimises the metric directly, using soft probabilities p_i so it stays differentiable.

Compound loss function

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}}$$

CE supplies dense, stable gradients early in training; Dice stops the model from collapsing to background. This is what nnU-Net uses, and it is a genuinely hard baseline to beat.

TRAINING LOOP, CONCRETELY

The standard training loop; only the model, loss and metric change:

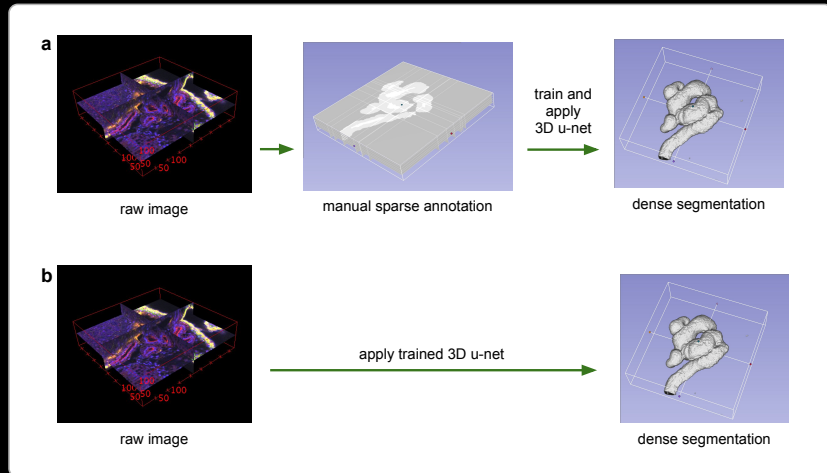
```
for x, mask in loader:
    logits = unet(x)
    p = logits.sigmoid()
    loss = bce_with_logits(logits, mask) \
        + dice_loss(p, mask)
    optimizer.zero_grad()
    loss.backward()
    optimizer.step()
# predict:  p > 0.5
```

- ▶ Output is a **per-pixel logit map**.
- ▶ Augment **image and mask together**: a flip that misses the mask silently corrupts the label.
- ▶ Validate on **Dice**, and keep the best-Dice checkpoint.

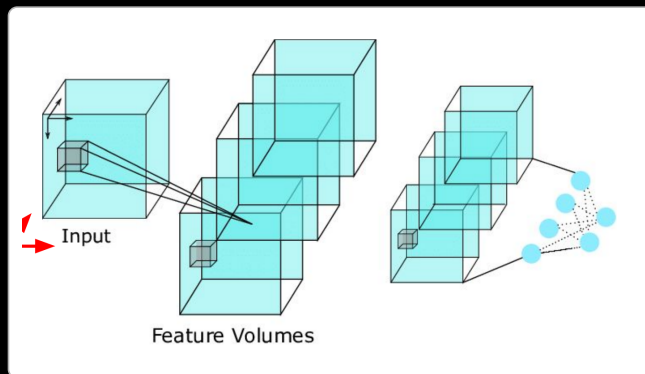
Going 3D

MEDICAL IMAGES ARE VOLUMES, NOT PICTURES

- ▶ Volumetric data is abundant in biomedical data analysis. Thus, annotation of large volumes in a slice-by-slice manner is **very tedious**.
- ▶ Annotation of such data with segmentation labels causes difficulties, since only 2D slices can be shown on a computer screen.



FROM 2D TO 3D CONVOLUTION



2D convolution

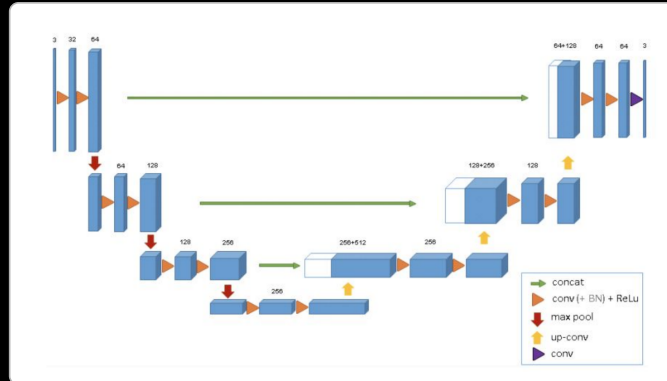
A $k \times k$ filter slides over height and width, and spans the *full* channel depth. Two directions of movement.

3D convolution

A $k \times k \times k$ filter slides along x , y and z . Three directions: depth is now a spatial axis, not a channel axis.

3D U-NET

While the u-net is an entirely 2D architecture, the network proposed in this paper takes 3D volumes as input and processes them with corresponding 3D operations, in particular, 3D convolutions, 3D max pooling, and 3D up-convolutional layers.



Analysis path

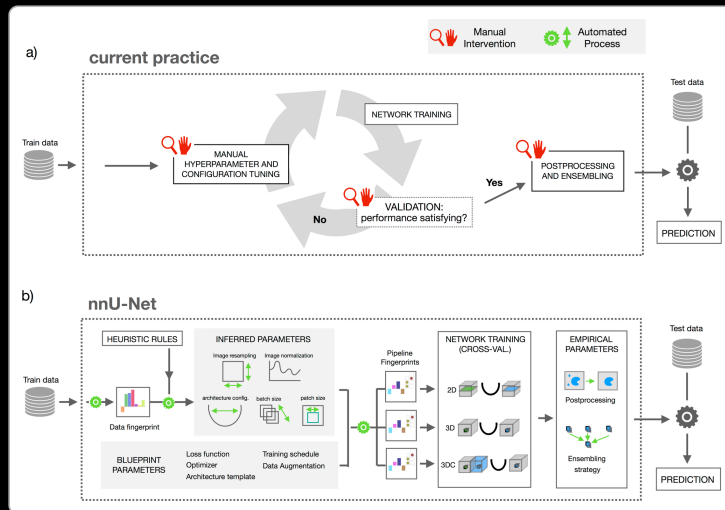
Four resolution steps. Each layer contains two $3 \times 3 \times 3$ convolutions each followed by a ReLU, and then a $2 \times 2 \times 2$ max pooling with strides of two in each dimension.

Synthesis path

Each layer is an upconvolution of $2 \times 2 \times 2$ by strides of two, followed by two $3 \times 3 \times 3$ convolutions each followed by a ReLU. Shortcut connections from layers of equal resolution in the analysis path provide the **essential high-resolution features**.

**nnU-Net: stop designing
architectures**

nnU-Net: CONFIGURE THE PIPELINE, NOT THE NETWORK



(a) current practice

A manual trial-and-error loop: tune, train, look at validation, tune again. The red hands are the human.

(b) nnU-Net

Fingerprint the data, run the rules, train 2D / 3D / cascade, pick the ensemble. **No hands**. It uses a plain U-Net, hence *no new net*.

WHAT nnU-NET ACTUALLY DECIDES

Fixed	same for every dataset	plain U-Net, loss = CE + Dice, SGD with Nesterov momentum, deep supervision (a loss at every decoder level), the augmentation set
Rule-based	computed from the fingerprint	patch size, batch size, network depth and downsampling, target spacing and resampling, intensity normalisation
Empirical	chosen by cross-validation	2D vs 3D vs 3D-cascade, which models to ensemble, whether to apply connected-component postprocessing

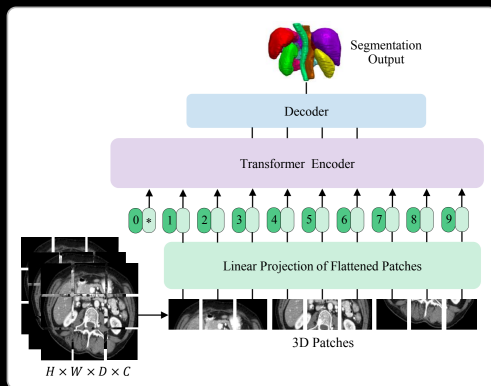
TRAINED FROM SCRATCH ON CHALLENGE DATA ALONE

33/53 new state of the art **23** datasets **11** challenges

Elsewhere, on par with or close to the top entry: organs, tumours and cells, in MRI, CT, electron and fluorescence microscopy.

Other transformers in medical image segmentation

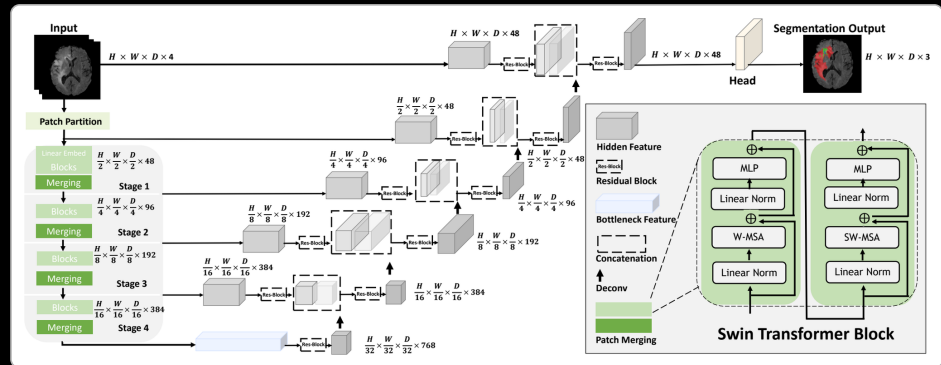
UNETR: A TRANSFORMER ENCODER



- ▶ Convolution is **local**.
- ▶ The volume becomes **3D patches**.
- ▶ A **transformer** encodes them.
- ▶ A CNN decoder rebuilds the mask.

Only the encoder changed.

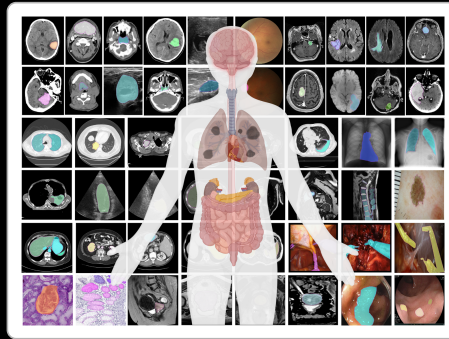
SWIN UNETR: ATTENTION IN SHIFTED WINDOWS



Attention inside **shifted windows**. Four stages, five resolutions, same CNN decoder.

Built for BraTS, where tumours vary in size.

OR SKIP TASK-SPECIFIC TRAINING: MEDSAM



- ▶ One model across **ten modalities**.
- ▶ Prompt it with a **bounding box**.
- ▶ **No fine tuning for your task.**

1.57M image-mask pairs

10 modalities

30+ cancer types

Expert annotation time, **82% less**.